

Báo cáo cancellation probability — theo giá trị biến phân loại

Tạo lúc: 08/07/2026 12:13

Mô hình: LightGBM v2 (Optuna tuned)

Phạm vi: 82,811 booking | Ngưỡng: 0.35

Đi kèm: data/11_cancellation_probability_scores.csv

1. Mục tiêu

Bảng mean P(hủy) theo từng giá trị biến phân loại (6 biến v2), so sánh với tỷ lệ hủy thực tế và tỷ lệ dự đoán Hủy @ 0,35.

Cột bảng	Ý nghĩa
n	Số booking
mean_P_huy	Trung bình xác suất hủy dự đoán
actual_cancel_%	Tỷ lệ is_canceled=1 thực tế
predicted_cancel_%	Tỷ lệ dự đoán Hủy @ 0,35

deposit_type

3 giá trị — sắp xếp theo mean P(hủy) giảm dần.

deposit_type	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
Non Refund	963	0.9533	95.0%	96.6%
No Deposit	81,767	0.3974	27.3%	50.8%
Refundable	81	0.2926	28.4%	30.9%

Top 3 rủi ro cao: Non Refund (0.953), No Deposit (0.397), Refundable (0.293)

Top 3 rủi ro thấp: Refundable (0.293), No Deposit (0.397), Non Refund (0.953)

market_segment

8 giá trị — sắp xếp theo mean P(hủy) giảm dần.

market_segment	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
Online TA	50,391	0.5032	35.5%	66.4%
Groups	3,690	0.3639	31.2%	42.5%

market_segment	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
Undefined	2	0.3079	100.0%	0.0%
Aviation	220	0.2821	19.1%	40.0%
Direct	11,351	0.2661	14.9%	27.8%
Complementary	619	0.2425	13.1%	20.2%
Corporate	3,678	0.2351	12.8%	20.8%
Offline TA/TO	12,860	0.2049	15.1%	26.1%

Top 3 rủi ro cao: Online TA (0.503), Groups (0.364), Undefined (0.308)

Top 3 rủi ro thấp: Offline TA/TO (0.205), Corporate (0.235), Complementary (0.242)

country

178 giá trị — sắp xếp theo mean P(hủy) giảm dần.

country	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
FJI	1	0.9493	100.0%	100.0%
NIC	1	0.8626	100.0%	100.0%
ARE	48	0.8593	85.4%	97.9%
FRO	3	0.8573	66.7%	100.0%
NAM	1	0.8370	0.0%	100.0%
BEN	3	0.8320	100.0%	100.0%
VGB	1	0.8254	100.0%	100.0%
KHM	1	0.8193	100.0%	100.0%
ATF	1	0.8158	0.0%	100.0%
IMN	2	0.8070	100.0%	100.0%
HND	1	0.7988	100.0%	100.0%
KIR	1	0.7916	0.0%	100.0%
GLP	1	0.7386	100.0%	100.0%
SAU	45	0.7235	66.7%	86.7%
JEY	8	0.7055	100.0%	87.5%
BGD	9	0.7020	77.8%	100.0%
TJK	2	0.6931	50.0%	100.0%
MYT	2	0.6611	100.0%	100.0%
BHR	5	0.6469	80.0%	100.0%

country	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
HKG	23	0.6468	91.3%	91.3%
RWA	2	0.6458	0.0%	100.0%
MRT	1	0.6390	0.0%	100.0%
AGO	339	0.6290	56.3%	66.1%
PAN	8	0.6231	0.0%	75.0%
GHA	4	0.6206	50.0%	100.0%
MAC	10	0.6196	90.0%	90.0%
GEO	19	0.6090	73.7%	73.7%
CHN	785	0.5938	49.4%	72.0%
PRY	4	0.5848	0.0%	75.0%
SEN	11	0.5838	72.7%	72.7%
QAT	14	0.5826	71.4%	78.6%
MAR	229	0.5482	45.4%	65.9%
PHL	31	0.5438	61.3%	74.2%
TUR	201	0.5419	47.8%	67.7%
SLE	1	0.5390	0.0%	100.0%
AND	7	0.5373	71.4%	85.7%
PAK	13	0.5274	61.5%	69.2%
RUS	545	0.5184	39.4%	65.5%
LUX	253	0.5161	39.1%	71.1%
TZA	5	0.5119	60.0%	100.0%
PRT	25,299	0.5072	36.8%	62.1%
AZE	16	0.5060	50.0%	68.8%
NCL	1	0.5044	0.0%	100.0%
MDV	12	0.4996	75.0%	58.3%
BRA	1,936	0.4965	36.8%	65.8%
KOR	118	0.4922	36.4%	66.1%
ITA	2,900	0.4910	36.1%	63.4%
KWT	16	0.4897	37.5%	62.5%
ABW	2	0.4893	0.0%	50.0%
IDN	25	0.4807	56.0%	56.0%
ZMB	2	0.4789	50.0%	100.0%

country	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
ASM	1	0.4776	0.0%	100.0%
TWN	41	0.4749	31.7%	65.9%
GIB	14	0.4726	78.6%	78.6%
ZAF	78	0.4703	37.2%	57.7%
HUN	199	0.4698	32.2%	63.3%
DOM	12	0.4678	50.0%	58.3%
GGY	2	0.4626	100.0%	50.0%
TMP	3	0.4581	33.3%	66.7%
UMI	1	0.4566	100.0%	100.0%
ISR	370	0.4541	35.4%	63.8%
CPV	23	0.4524	52.2%	60.9%
ETH	3	0.4432	33.3%	66.7%
MLT	17	0.4398	29.4%	58.8%
THA	38	0.4390	36.8%	55.3%
SGP	34	0.4386	35.3%	61.8%
URY	28	0.4342	32.1%	71.4%
DMA	1	0.4332	0.0%	100.0%
NOR	491	0.4324	28.7%	54.8%
SVK	48	0.4321	20.8%	56.2%
TUN	37	0.4273	48.6%	62.2%
MUS	7	0.4244	14.3%	71.4%
UGA	2	0.4198	0.0%	50.0%
PLW	1	0.4196	0.0%	100.0%
BLR	24	0.4191	33.3%	54.2%
COM	2	0.4179	0.0%	50.0%
BIH	11	0.4114	27.3%	54.5%
IRN	77	0.4083	27.3%	61.0%
ARG	199	0.4037	25.6%	53.8%
SLV	2	0.3999	0.0%	50.0%
EST	77	0.3987	23.4%	51.9%
COL	68	0.3978	33.8%	55.9%
CHE	1,545	0.3955	24.8%	54.8%

country	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
CZE	128	0.3952	25.8%	53.1%
UKR	60	0.3895	33.3%	51.7%
ROU	447	0.3891	26.2%	51.7%
HRV	91	0.3883	25.3%	53.8%
NGA	30	0.3875	56.7%	50.0%
IND	138	0.3869	23.9%	51.4%
GAB	4	0.3867	50.0%	50.0%
EGY	30	0.3857	33.3%	56.7%
AUS	370	0.3846	24.6%	50.8%
ESP	6,973	0.3829	26.2%	50.2%
CIV	6	0.3822	33.3%	50.0%
USA	1,825	0.3799	24.8%	50.3%
GRC	114	0.3769	27.2%	50.0%
VEN	21	0.3759	38.1%	47.6%
MNE	5	0.3750	40.0%	40.0%
LIE	3	0.3743	33.3%	66.7%
ARM	8	0.3702	25.0%	50.0%
DNK	373	0.3700	24.4%	46.1%
ISL	22	0.3687	18.2%	50.0%
KAZ	19	0.3657	26.3%	47.4%
MKD	3	0.3652	33.3%	33.3%
UZB	4	0.3622	50.0%	50.0%
VNM	8	0.3618	25.0%	50.0%
KEN	6	0.3611	33.3%	16.7%
MCO	4	0.3568	25.0%	50.0%
SVN	44	0.3561	20.5%	43.2%
SUR	5	0.3509	0.0%	40.0%
MMR	1	0.3504	0.0%	100.0%
IRL	2,893	0.3502	22.2%	46.6%
MOZ	50	0.3447	32.0%	42.0%
SWE	783	0.3391	22.2%	42.3%
CMR	10	0.3381	0.0%	50.0%

country	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
ZWE	4	0.3380	50.0%	50.0%
CYP	40	0.3374	20.0%	45.0%
DZA	78	0.3365	26.9%	41.0%
CHL	63	0.3293	22.2%	41.3%
POL	739	0.3280	22.1%	42.4%
LBN	28	0.3229	21.4%	42.9%
BEL	2,022	0.3229	20.2%	38.0%
FRA	8,464	0.3188	20.1%	41.9%
CN	1,055	0.3180	21.1%	42.0%
ECU	24	0.3160	29.2%	41.7%
LVA	47	0.3155	17.0%	42.6%
DEU	5,109	0.3135	20.2%	41.8%
SYC	2	0.3127	50.0%	50.0%
NLD	1,864	0.3092	18.6%	37.6%
BGR	67	0.3067	13.4%	40.3%
GBR	9,921	0.2970	19.6%	39.7%
MYS	24	0.2947	12.5%	41.7%
AUT	887	0.2936	17.9%	36.3%
JOR	18	0.2912	16.7%	27.8%
ATA	2	0.2898	0.0%	50.0%
CUB	8	0.2846	0.0%	37.5%
MEX	71	0.2830	12.7%	33.8%
CRI	18	0.2808	5.6%	33.3%
FIN	416	0.2686	15.4%	27.4%
PER	25	0.2637	24.0%	32.0%
ALB	11	0.2602	18.2%	36.4%
JPN	176	0.2514	13.6%	23.3%
OMN	18	0.2453	22.2%	33.3%
SRB	47	0.2309	6.4%	34.0%
GTM	3	0.2041	0.0%	33.3%
TGO	2	0.2001	0.0%	50.0%
BOL	3	0.1997	0.0%	0.0%

country	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
LTU	73	0.1980	9.6%	21.9%
BFA	1	0.1926	0.0%	0.0%
LKA	5	0.1899	0.0%	20.0%
NZL	63	0.1874	6.3%	20.6%
JAM	6	0.1854	0.0%	33.3%
BWA	1	0.1647	0.0%	0.0%
LAO	2	0.1636	0.0%	0.0%
IRQ	10	0.1484	0.0%	20.0%
GNB	9	0.1472	11.1%	11.1%
SYR	3	0.1441	0.0%	33.3%
AIA	1	0.1372	0.0%	0.0%
Unknown	437	0.1354	7.6%	9.4%
PRI	11	0.1079	9.1%	9.1%
MLI	1	0.0973	0.0%	0.0%
CAF	5	0.0929	0.0%	0.0%
DJI	1	0.0853	0.0%	0.0%
NPL	1	0.0761	0.0%	0.0%
SMR	1	0.0685	0.0%	0.0%
STP	2	0.0670	0.0%	0.0%
CYM	1	0.0642	0.0%	0.0%
LBY	7	0.0634	0.0%	0.0%
MWI	2	0.0528	0.0%	0.0%
KNA	2	0.0398	0.0%	0.0%
PYF	1	0.0372	0.0%	0.0%
MDG	1	0.0257	0.0%	0.0%
GUY	1	0.0257	0.0%	0.0%
SDN	1	0.0253	0.0%	0.0%
LCA	1	0.0252	0.0%	0.0%
BRB	1	0.0226	0.0%	0.0%
BDI	1	0.0224	0.0%	0.0%
BHS	1	0.0104	0.0%	0.0%

Top 3 rủi ro cao: FJI (0.949), NIC (0.863), ARE (0.859)
Top 3 rủi ro thấp: BHS (0.010), BDI (0.022), BRB (0.023)

distribution_channel

5 giá trị — sắp xếp theo mean P(hủy) giảm dần.

distribution_channel	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
TA/TO	65,956	0.4408	31.5%	57.7%
Undefined	5	0.3804	80.0%	60.0%
GDS	172	0.3345	19.8%	44.2%
Direct	12,291	0.2663	15.1%	27.6%
Corporate	4,387	0.2343	13.6%	22.2%

Top 3 rủi ro cao: TA/TO (0.441), Undefined (0.380), GDS (0.335)
Top 3 rủi ro thấp: Corporate (0.234), Direct (0.266), GDS (0.335)

customer_type

4 giá trị — sắp xếp theo mean P(hủy) giảm dần.

customer_type	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
Transient	69,939	0.4369	30.4%	56.1%
Transient-Party	9,294	0.2339	15.8%	26.9%
Contract	3,068	0.2031	16.5%	23.8%
Group	510	0.1554	10.6%	14.3%

Top 3 rủi ro cao: Transient (0.437), Transient-Party (0.234), Contract (0.203)
Top 3 rủi ro thấp: Group (0.155), Contract (0.203), Transient-Party (0.234)

hotel

2 giá trị — sắp xếp theo mean P(hủy) giảm dần.

hotel	n	mean P(hủy)	Hủy thực tế %	Dự đoán Hủy %
City Hotel	50,686	0.4383	30.7%	57.3%
Resort Hotel	32,125	0.3491	24.1%	42.0%

Top 3 rủi ro cao: City Hotel (0.438), Resort Hotel (0.349)
Top 3 rủi ro thấp: Resort Hotel (0.349), City Hotel (0.438)

Ghi chú

- Mean P(hủy) theo nhóm **không đồng nghĩa** xác suất nhân quả — chỉ phản ánh scoring trung bình của mô hình trên segment đó.
- Biến `country` có nhiều mức hiếm; các nước ít booking có mean P(hủy) biến động mạnh.
- So sánh `actual_cancel_%` vs `mean_P_huy` giúp đánh giá calibration theo segment.

Sinh tự động bởi `_run_11_probability_reports.py`.